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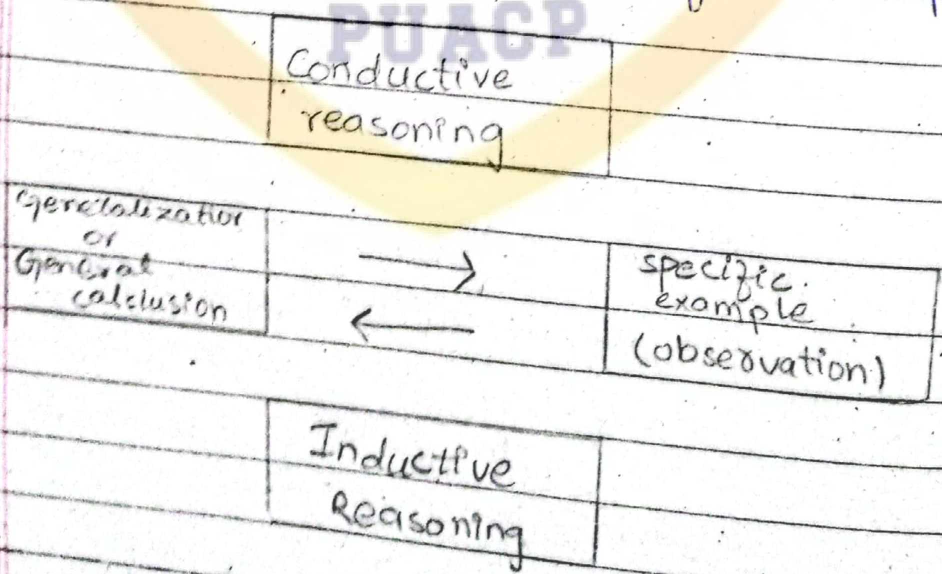
Quantitive Reasoning

→ **Inductive reasoning:**
Inductive reasoning is making generalization from observed example.

→ **Conductive reasoning:**
Conductive reasoning is process of applying a given generalization ~~or~~ rule to specific example.



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Number System

→ Decimal number System:

The Decimal number system is a number system of base or radix equal to 10.

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

→ Binary number System:

The binary number system is a number system of base or radix equal to 2. There are two symbols used to represent number : 0 and 1.

→ Octal number System:

The octal number system is used extensively in digital work because it is easy to convert from octal to binary, vice versa. The octal number system has a base or radix equal to 8.

0, 1, 2, 3, 4, 5, 7

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Examples

1- Convert binary to decimal:

(a) 1101

$1101 = (1\times2^3) + (1\times2^2) + (0\times2^1) + (1\times2^0)$

$8 + 4 + 1 = 13$

2- Convert binary to decimal:

0.011

$0.011 = (0\times2^{-1}) + (1\times2^{-2}) + (1\times2^{-3})$

$= 0 + 1/4 + 1/8$

$= 0.25 + 0.125 = 0.375$

3- Convert binary to decimal:

110.011

$= (1\times2^2) + (1\times2^1) + (0\times2^0) + (0\times2^{-1}) + (1\times2^{-2}) + (1\times2)^{-3}$

$= 4 + 2 + 1/4 + 1/8$

$= 4 + 2 + 0.25 + 0.125 = 6.375$

4- Convert decimal to binary:

17

2	17
2	8 – 1
2	4 – 0
2	2 – 0
2	1 – 0
	0 – 1

17 = 10001

0.625

$0.625 \times 2 = 1.250$

$0.250 \times 2 = 0.500$

$0.500 \times 2 = 1.000$

$0.625 = 0.101$

(c) 0.6

$0.6 \times 2 = 1.2$

$1.2 \times 2 = 0.4$

$0.4 \times 2 = 0.8$ $\quad$ $0.6 = 0.10011$

$0.8 \times 2 = 1.6$

$0.6 \times 2 = 1.2$

$0.2 \times 2 =$

8- Convert octal numbers to decimal binary.

$(35)_8$

$= (3\times8)^1 + (5\times8)^0$

$= 24 + 5$

$= 29$

$(100)_8$

$(1\times8)^2 + (0\times8)^1 (0\times8)^0$

$(64) + 0 + 0$

72 64

$(0.24)_8$

$(2\times8)^{-1} + (4\times8)^{-2}$

$2/4 + 4/8$

0.3125

9- Convert decimal number to octal:

8	245
8	30 - 5
8	3 - 6
	0 - 3

$(245)_{10} = (365)_8$

8	175
8	21 - 7
8	2 - 5
	0 - 2

$(175)_{10} = (257)_8$

10 - Convert decimal fraction to octal:

0.432

$0.432 \times 8 = 3.456$

$0.456 \times 8 = 3.648$ $\quad$ $(0.432)_{10} = (0.3351)_8$

$0.648 \times 8 = 5.184$

$0.184 \times 8 = 1.472$

11- Convert octal to binary:

$(501)_8$

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octal	binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111
8	
9	

5 = 101
0 = 000
1 = 001

$(501)_8 = 101000001$

11- Convert 11010101.01101 to an octal number.

011 = 3
010 = 2
101 = 5
011 = 3
10 = 2

$(11010101.01101)_{10} = (325.32)_8$

12- Add 1010, 1001 and 1101.

```
 1010
 1001
 1101
-----
10000
```

Sets

Natural set :

$N = 1, 2, 3, 4 \ldots$

whole set :

$W = 0, 1, 2 \ldots$

Integers

$Z = 0, \pm 1, \pm 2, \pm 3 \ldots$

Real set :

$\frac{P}{Q}$ and $Q \neq 0$

- **Rational** terminating
- **Irrational** non terminating
  - Recurring
  - non-recurring

Complex number :

$a + ib$

real imaginary

Data and its types :

- Data can be define as collection of facts, or information from which conclusion can be drawn.
- Data are measurements or observations that are collected as a rouse source of information.
- Data is a set of values of subjects w.r.t Qualitative or quantitative.
- Data may be devided as primary data and secondary data.

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* **primary data** is collected by ? and is used by a collected.

As primary data is collected directly from its sources, therefore it is also considered more accurate because it was gathered personaly.

* **Secondary data:**

Secondary data is collected by someone else or othere then person analyzing it. It was collected by experts for different purposes but it is being used for research now.

Both primary and secondary data may be qualitative or quantitative.

Types of data:

Data come is in general is of two types.

(i) qualitative or catagorical.

(ii) quantitative or numerical.

*1 **Qualitative Data:**

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Qualitative data can take on one of a limited, and usually fixed, number of possible values.

Example:

Name of months
Name of days in weak.
Blood group of persons.
Gender
colur of eyes, hair etc.

→ Qualitative or catogorical data is further devided into following two types:

- Nominal
- Ordinal

I Nominal:

Nominal data describe catagories or groups that do not have a specific order of variables.

Example:

Colours of cars.
Blood types of human being.
Gender.

II Ordinal data:

Ordinal variables have two or more catagories/groups that can be ordered or ranked.

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Example:

- Responce data that ranges from a strongly disagree to a strongly agree.
- First, second, third and so on....
  Latter grades: A, B, C ...

Nominal data is further devided into following two types.

- Polytomous
- Dichotomous / binary variables.

(a) → **Dichotomous:** Dichotomous variable can take exactly two values.

→ yes, no,
→ True, false
on, OFF

(b) → **polytomous:** catagorical variable can with more than two possible values are called polytomous variables.

colours of cards, Blood groups.

★2 Quantitative Data:

A numerical or Quatitative variable is a data variable that takes on any value within a finite or infinite interval.

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Example:

height, weight, age, number of chairs in room, world population.

Quantitative data is further devided into following two catagories:

- discrete and
- continuos.

I Discrete :

Discrete data can take any numerical value, but are countable in a finite interval of time.

Example:

number of students in a class.

worlds population.

Not: It might take a long time to count but the fact is it is still countable.

Continuous:

continuous variable can take any numerical value but can't be countable.

It would take forever to count.

Example:

The precise age of person can be: 25 years, 10 month, 2 days, 5 hours, 40 minutes, 4 sec, 5 msec, 8 nanosec, ...

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- Data
  - catagorical / Quanlitative
    - Nominal
      - Dichotonous
      - polytomous
    - ordinal
  - Numerical / Quantitative
    - Discrete
    - Continuos

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Nominal Scale:

It does not imply and Quantitative value or order.

2- Ordinal Scale:

It is used to represent non mathematical ideas. such as degree of pain, va Lavel of satisfaction, level of happiness.

Interval scale:

It is define as a numerical scale, where the order of variables as well as the difference between these variables is known ratio scale.

Ratio Scale:

It is a variable measurement Scale. that not only produces order of variable but also make the difference b/w non variables along with information about the true zero.

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→1 Tabular and graphical representation of data:

- **Data:** A set of values recorded for an event is called data. Data can be stored and presented in various ways to draw some conclusion / inference. Data can be classifies as:
primary data, Secondary, Qualitative, Quantitative etc.
A data presented without any oder, does not allow to derive any conclusion from it therefore it is essential to organize data. This can be done by summasing data into frequency an distributed table. The main objectives of data classification is:

- To make a proper use of raw informations / data.
- To study the data and make comparison easier.
- To use the collected material for satistical treatment.
- To simplify the complexities of raw data.
- To draw statistical (conclusions) inferences from the raw data.
- To keep the unnessessary information aside.

Frequency distribution :

or a frequency table is the tabular arrangement of data by classes. together with the corresponding class frequencies.

The main purpose of frequency distribution is to organize data into a more compact form without (غیر واضح) obscuring) essential information contained in the values.

Example : Height of 15 plants in inchese is recorded as follow :

53, 48, 55, 51, 50, 57, 54, 56, 56, 54, 53, 52, 53, 49, 53.

Class	frequency	Relative(f)	Cummutative(f)
48 - 50	2		
50 - 52	2		
52 - 54	5		
54 - 56	3		
56 - 58	3		

incomplete

Construction of group frequency table :

Important points to be Considered at the time of construction of frequency distribution tables are no. of classes, class interval, class limit and mid value / point.

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(i) No. of classes:
No. of classes is an important factor for prepairing frequency table. Their is no fixed rule that how many classes are to be taken. Generally depends upon observation of available data, min three classes and maximium 20 classes are formed. The size of class interval also depends upon range of data and no. of classes. It is equal to the difference b/w heighest and lowest value.

(ii) Class interval:
It depends upon the range of data and no. of classes. It is given by $i = \frac{H - L}{C}$ where i is class interval H is heighest value and L is lowest value and c is no. of classes.

(iii) class Limit

included in the class, which are In a class 20 - 25 lowest value is and highest value is 25.

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Mid value / Mid point:
The central point of a class interval is mid point. It can be calculated by adding the upper and lower limits of a class, and then deviding the sum by '2' i.e.
Mid point equal $= \frac{L_1 + L_2}{2}$, where L_1 is lower limit, and L_2 is upper limit of a class.

Types of frequency tables:

Overlapping frequency distribution table:
Values of variables are grouped in such a way that the upper limit of one class interval is represented in the next class interval.

non-overlapping frequency distribution table:
Values of variables are grouped in such a way that the upper level of one class interval do not overlap the preceding class interval.

Example: For the number of plants regarding ~~their~~ heights are then ranging from 15 - 28 the class intervals may be such as: 15 - 17, 18 - 20, 21 - 23, ...
This is non-overlapping frequency distribution table

Methods

There are two main methords of representation of statiscal form of data.

- Table method
- graph methord.

1- Table Methord:

Following are the features of tabular representation of statistical data:

(i) Tabulation is a process of orderly arrangement of data into rows or columns or series where they can be read at a glance.

(ii) This process is called summerization of data in an orderly manner within a limited space. It helps in

(iii) It helps in simplifying the raw data.

(iv) comparison can be made easily. (واضح/کرنا)

(v) It reveals the (نمونہ) pattern (تقسیم) of distribution of attributes, defacts, omitions and errors. It accurate figures are given that tabular representation has a great value for experts.

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Types of table:

I Simple table : in this type of table only one parameter is considered. Example : length of papaya plant in a feild.

Length of plant	6 – 10	11 – 15	16 – 20
No. of plants	6	9	11

II Complex table :

In this type of table more than one parameter is considered. Example : length, Gender and Health (healthy or infected) of a plant.

Length of Plant cm	Infected male	Healthy male	Infected Female	Healthy Female
6 – 10	2	1	1	1
11 – 15	2	4	2	2
16 – 20	1	4	4	4

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Graphical representation of data.

→ Graph:

- A graph is a pictorial presentation of relationship between variables espacially to express the change in some quantity over a period of time.

- Graph is a visual form of the representation of statistical data.

- Graphical method enables statistician to present quantitative data in a simple, clear and effective manner.

- Comparisons can be easily made between two or more phenomena with the help of graph.

- To obtain clear picture we can represent frequency table pictorically. Such a visual Pictorial representation can be done through graphs.

Purpose of graph:

- To compare two or more numbers: The comparison is often by bars of different lengths.
- To express the distribusion of indivisual objects of measurements

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into different catagories. The frequency distribution of numerical catagories is usally represented by histogram.

- The distribution of indivisuals into non-numerical catagories can be shown as a bar-diagram. The length of bars represents the number of observations (or frequency) in each catagory.

- If the frequencies are expressed as percentages, tataling 100%, a convenient way is a pie chart.

→ Types of graph

- (4) Line graph
- (5) Bar graph — daigram nhi × kri %age kri hai multiple
- (6) pie graph
- Histogram (1) merets demerets major steps fre
- daigram merets demerets (2) frequency polygon
- (3) frequency curve

scratter bar daigram merets demerets figure

1. Histograms:

- This is one of the most popular methods for displaying the frequency distribution.
- In this type of representation, the given data is plotted in the form of series of rectangles.

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- The height of a rectangle is proportional to the repective frequency and width represents the class interval.
- The class intervals are marked along x-axis and the frequencies along y-axis, any blank spaces between the rectangles would mean that the category is empty and there are no values in that class interval.
- A histogram is two-dimensional in which both the length and the width are important.

→ Merits of histogram:

1- Histogram ~~ca~~ gives the idea about the amount of variability present in the data.

2- It is useful to find out mode.

3

→ Demerits of Histogram:

1- Histogram can be drawn for frequency distribution with open-end class.

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2- Histogram is not a convenient method for comparisons especially the super-imposed histograms are usually confusing.

→ Major steps involved in construction of histogram:

- Arrange the data in ascending order.
- Find out class interval.
- prepare the frequency distribution diagram.
- Draw the histogram by taking class value on x-axis and frequency on y-axis.

2. Frequency polygon:

- It is a line chart of frequency distribution in which mid points of class intervals are plotted are joined by straight line.
- It is the variation of histogram is which instead of rectangles erect over the intervals, the points are plotted at the mid points of the tops of the corresponding ~~researches~~ rectangles in a histogram, and the successive points are joined by straight lines.
- frequency polygon is used in cases of time series, that is when the distribution of the variable is given by as a function of time.

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- E.g. Growth of plant over a period of time, trends in food production etc.

→ Merits:

1. It can be constructed quickly than histogram.
2. It enables to understand the pattern on the data more clearly than histogram.

→ Demerits:

It can not give an accurate picture as that given by histogram because in frequency polygon the areas above the various intervals are not exactly proportional to the frequencies

3. Frequency Curve:

- When the total frequency is large, and the class intervals are narrow so the frequency polygon or histogram will approach more and more towards the form of smooth curve. Such a smooth curve is called frequency curve.

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- Frequency curve is also called smoothed frequency daigram.

- In this, total area under the curve is equal to the area under the original histogram or polygon.
- This usually has single hump or mode (value with highest frequency.)

→ Scatter or Dot diagram: